

Relational Algebra

Definitions, Operators, and Examples

Introduction

Relational Algebra is a procedural query language that works on relations (tables) in a database. It consists of a set of operations that take one or two relations as input and produce a new relation as output. These operations form the theoretical foundation of relational databases and SQL.

This document covers the fundamental operations, their formal definitions, and examples.

Sample Relations

We will use the following example relations throughout the notes:

Employee(EmpID, Name, Dept, Salary)

EmpID	Name	Dept	Salary
101	Alice	HR	5000
102	Bob	IT	6000
103	Carol	IT	5500
104	Dave	Sales	4500

Department(DeptID, DeptName, Manager)

DeptID	DeptName	Manager
1	HR	Alice
2	IT	Bob
3	Sales	Eve

1. Selection (σ)

Definition

Selection operator $\sigma_{condition}(R)$ selects tuples from relation R that satisfy the given condition.

$$\sigma_{condition}(R) = \{t \in R \mid t \text{ satisfies } condition\}$$

Result: Relation with tuples filtered by the condition.

Example: Select employees from the IT department.

$$\sigma_{Dept='IT'}(Employee)$$

Result

EmpID	Name	Dept	Salary
102	Bob	IT	6000
103	Carol	IT	5500

2. Projection (π)

Definition

Projection operator $\pi_{attributes}(R)$ extracts specified columns (attributes) from relation R , removing duplicates.

$$\pi_{A_1, A_2, \dots, A_n}(R) = \{t[A_1, A_2, \dots, A_n] \mid t \in R\}$$

Result: Relation with only specified attributes.

Example: List the names and departments of all employees.

$$\pi_{Name, Dept}(Employee)$$

Result

Name	Dept
Alice	HR
Bob	IT
Carol	IT
Dave	Sales

3. Union ($\cup$)

Definition

Union operator $R \cup S$ combines tuples from relations R and S .

$$R \cup S = \{t \mid t \in R \text{ or } t \in S\}$$

Requirement: R and S must be union-compatible (same attributes & domains).

Example: Suppose we have two relations of employees on leave:

- $Leave1 = \{101, 104\}$
- $Leave2 = \{102, 105\}$

Then,

$$Leave1 \cup Leave2 = \{101, 102, 104, 105\}$$

4. Set Difference ($-$)

Definition

Set difference operator $R - S$ gives tuples in R but not in S .

$$R - S = \{t \mid t \in R \text{ and } t \notin S\}$$

Requirement: R and S must be union-compatible.

Example: Employees who are not on leave in $Leave2$.

$$Leave1 - Leave2 = \{101, 104\} - \{102, 105\} = \{101, 104\}$$

5. Cartesian Product ($\times$)

Definition

Cartesian product $R \times S$ combines every tuple of R with every tuple of S .

$$R \times S = \{t_r \circ t_s \mid t_r \in R, t_s \in S\}$$

Here, $t_r \circ t_s$ denotes concatenation of tuples.

Example: Cartesian product of

Employee(EmpID, Name), Department(DeptID, DeptName)
pairs every employee with every department.

6. Rename (ρ)

Definition

Rename operator $\rho_{NewName}(R)$ renames relation R or its attributes.

$$\rho_S(R) \quad \text{or} \quad \rho_{S(A,B)}(R)$$

This is useful to avoid ambiguity in operations like joins.

Example: Rename relation **Employee** to **Emp**:

$$\rho_{Emp}(\text{Employee})$$

7. Join Operations

Join combines tuples from two relations based on a condition.

a) Natural Join ($\bowtie$)

Definition

Natural join automatically joins two relations over all common attributes.

$$R \bowtie S = \{t_r \circ t_s \mid t_r \in R, t_s \in S, t_r[A] = t_s[A] \text{ for all common attributes } A\}$$

Example: Natural join of **Employee** and **Department** on attribute **Dept = DeptName** (assuming **Dept** in **Employee** corresponds to **DeptName** in **Department**):

$$\text{Employee} \bowtie_{\text{Employee.Dept} = \text{Department.DeptName}} \text{Department}$$

Result:

EmpID	Name	Dept	Salary	Manager
101	Alice	HR	5000	Alice
102	Bob	IT	6000	Bob
103	Carol	IT	5500	Bob
104	Dave	Sales	4500	Eve

b) Theta Join ($\bowtie_\theta$)

Definition

Theta join uses a general condition θ for joining two relations.

$$R \bowtie_\theta S = \sigma_\theta(R \times S)$$

Example: Join **Employee** and **Department** where $\text{Employee.Dept} = \text{Department.DeptName}$:

$$\text{Employee} \bowtie_{\text{Employee.Dept} = \text{Department.DeptName}} \text{Department}$$

8. Division ($\div$)

Definition

Division operator $R \div S$ returns tuples in R that match all tuples in S on the divisor attributes.

$$R(X, Y) \div S(Y) = \{x \mid \forall y \in S, (x, y) \in R\}$$

Use case: Find entities related to *all* entities in another set.

Example: Suppose relation

$$\text{WorksOn}(\text{EmpID}, \text{ProjectID})$$

and

$$\text{Projects}(\text{ProjectID})$$

Query: Find employees who work on all projects.

$$\text{WorksOn} \div \text{Projects}$$

Summary Table of Operators

Operator	Description	Symbol/Notation
Selection	Filter rows by condition	$\sigma_{condition}(R)$
Projection	Select specific columns	$\pi_{attributes}(R)$
Union	Tuples in R or S	$R \cup S$
Set Difference	Tuples in R not in S	$R - S$
Cartesian Product	Combine tuples of R and S	$R \times S$
Rename	Rename relation or attributes	$\rho_{NewName}(R)$
Natural Join	Join on common attributes	$R \bowtie S$
Theta Join	Join with condition θ	$R \bowtie_{\theta} S$
Division	Find tuples related to all in S	$R \div S$